DATASET PROFILING & SUMMARY STATS REPORT

DATASET USED: ADULT PSYCHIATRIC MORBIDITY SURVEY (APMS) 2014

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Introduction:

The dataset (APMS – Annual Status of Education Report type data) contains structured information collected from surveys. It includes multiple fields such as demographic details, education-related indicators, and categorical/numerical variables.

- the data is stored in csv (comma-separated values) format
- each row represents a record/observation
- each column represents a feature/attribute
- the dataset may include:
  o numeric data (e.g., age, scores)
  o categorical data (e.g., gender, category)
  o missing values (empty or null fields)

TECHNOLOGY USED

- **Python** – Programming language for analysis
- **Jupyter Notebook** – Interactive environment for coding and visualization
- **Pandas** – Data manipulation and analysis library

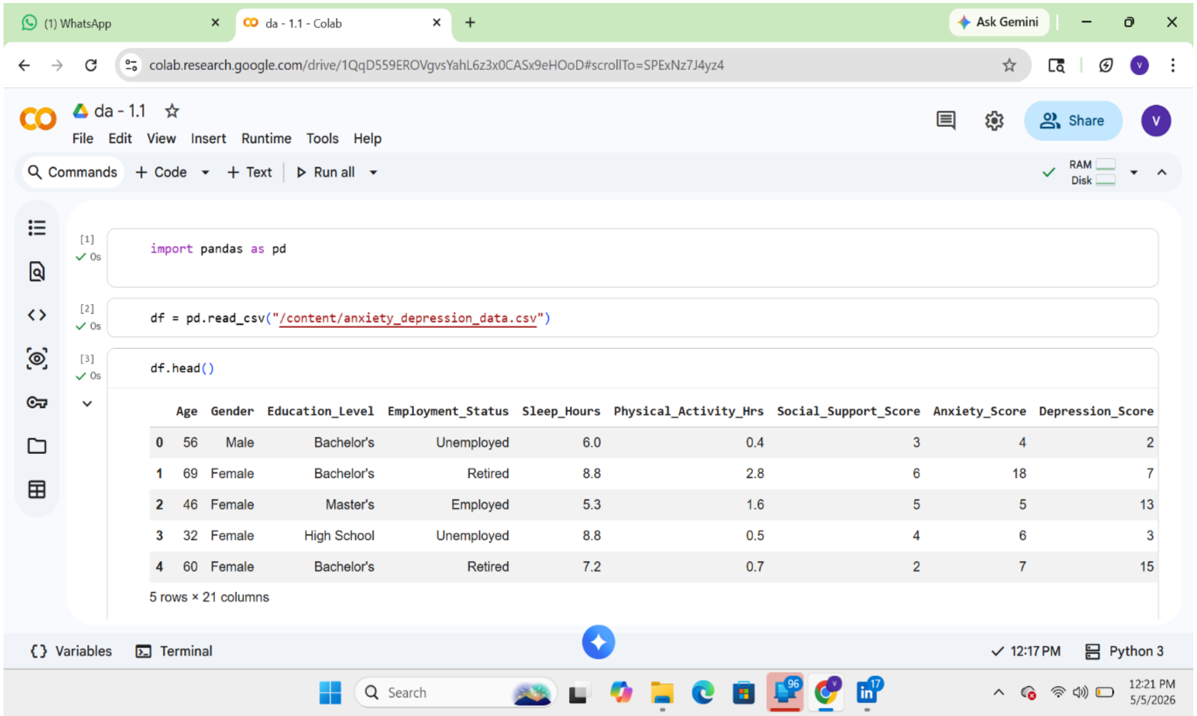
DATASET PROFILING:

Dataset profiling is the process of examining the dataset to understand its quality and characteristics.

Row Counts:

- Total number of records in the dataset

- Helps understand dataset size and completeness



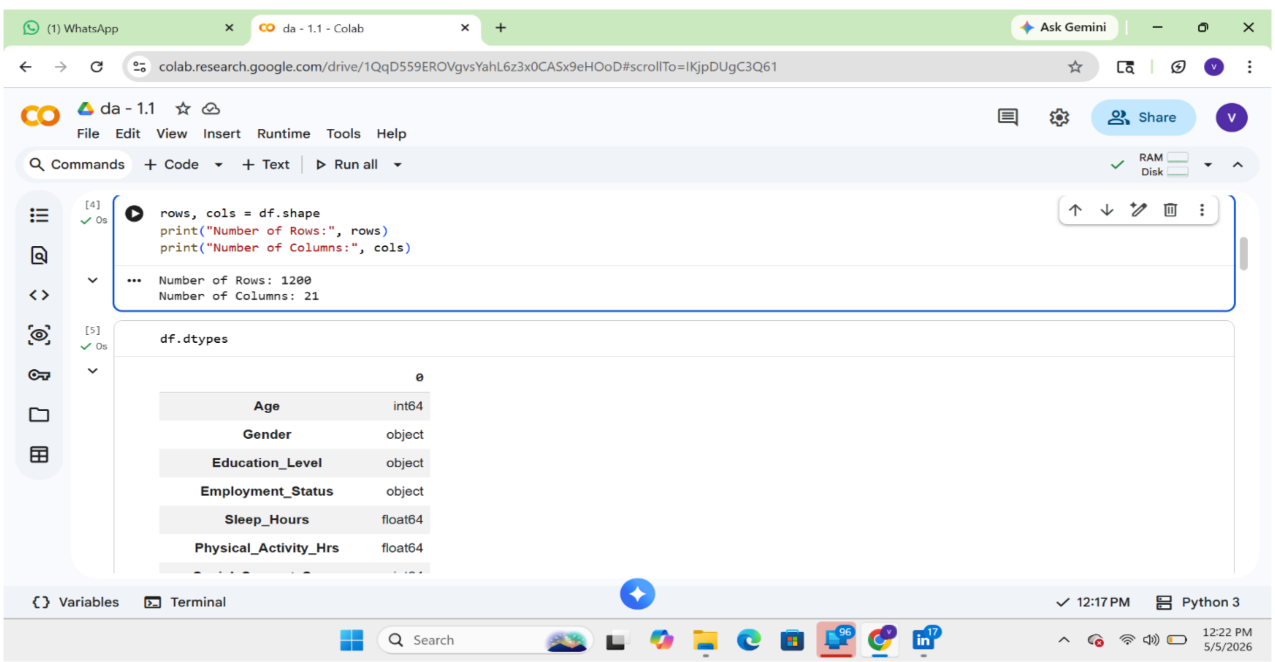
```python
import pandas as pd
```

```python
df = pd.read_csv("/content/anxiety_depression_data.csv")
```

```python
df.head()
```

	Age	Gender	Education_Level	Employment_Status	Sleep_Hours	Physical_Activity_Hrs	Social_Support_Score	Anxiety_Score	Depression_Score
0	56	Male	Bachelor's	Unemployed	6.0	0.4	3	4	2
1	69	Female	Bachelor's	Retired	8.8	2.8	6	18	7
2	46	Female	Master's	Employed	5.3	1.6	5	5	13
3	32	Female	High School	Unemployed	8.8	0.5	4	6	3
4	60	Female	Bachelor's	Retired	7.2	0.7	2	7	15

5 rows × 21 columns



```python
rows, cols = df.shape
print("Number of Rows:", rows)
print("Number of Columns:", cols)
```

```
Number of Rows: 1200
Number of Columns: 21
```

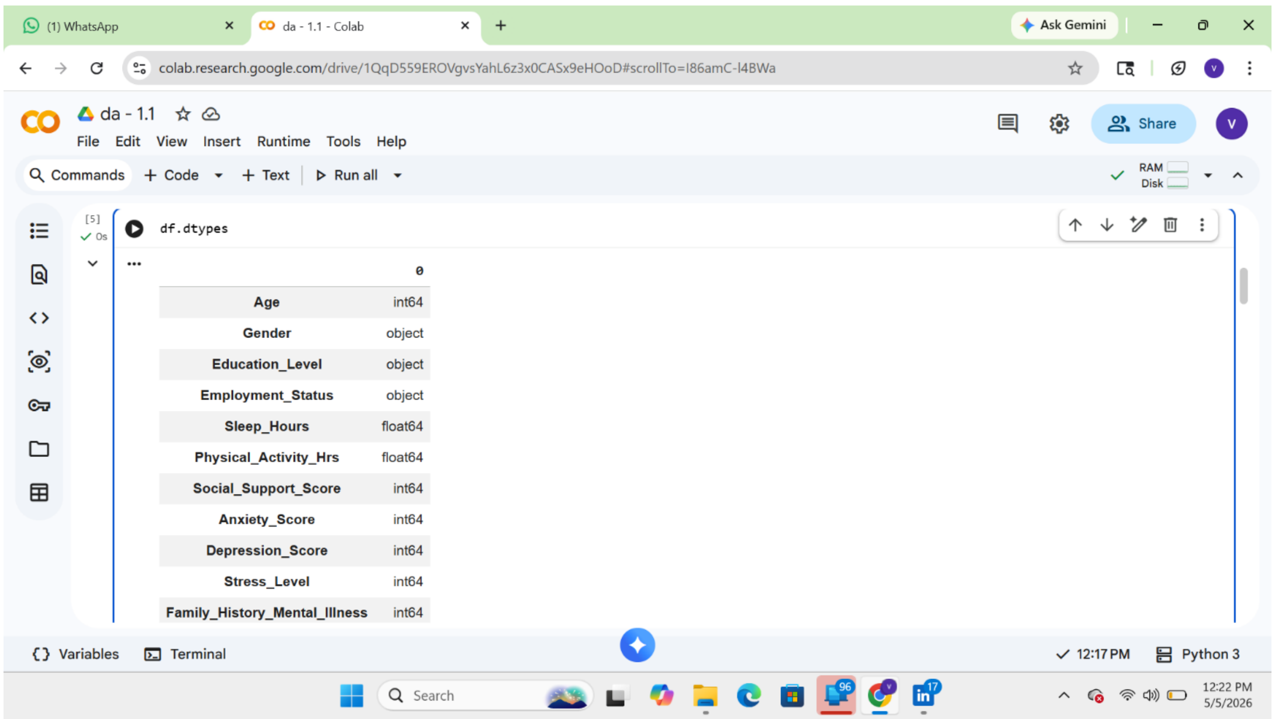
```python
df.dtypes
```

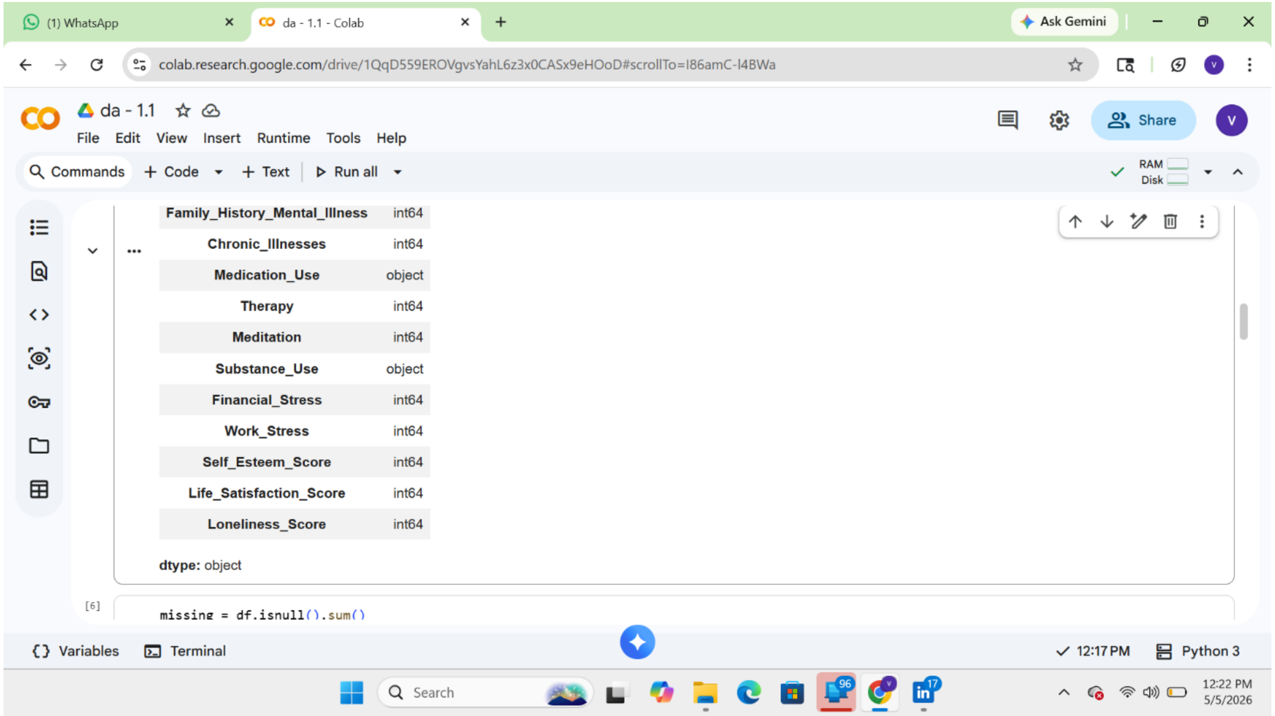
	0
Age	int64
Gender	object
Education_Level	object
Employment_Status	object
Sleep_Hours	float64
Physical_Activity_Hrs	float64

FIELD DATATYPES:

- Identifies whether columns are:

    1. Numerical (int, float)
    2. Categorical (object/string)



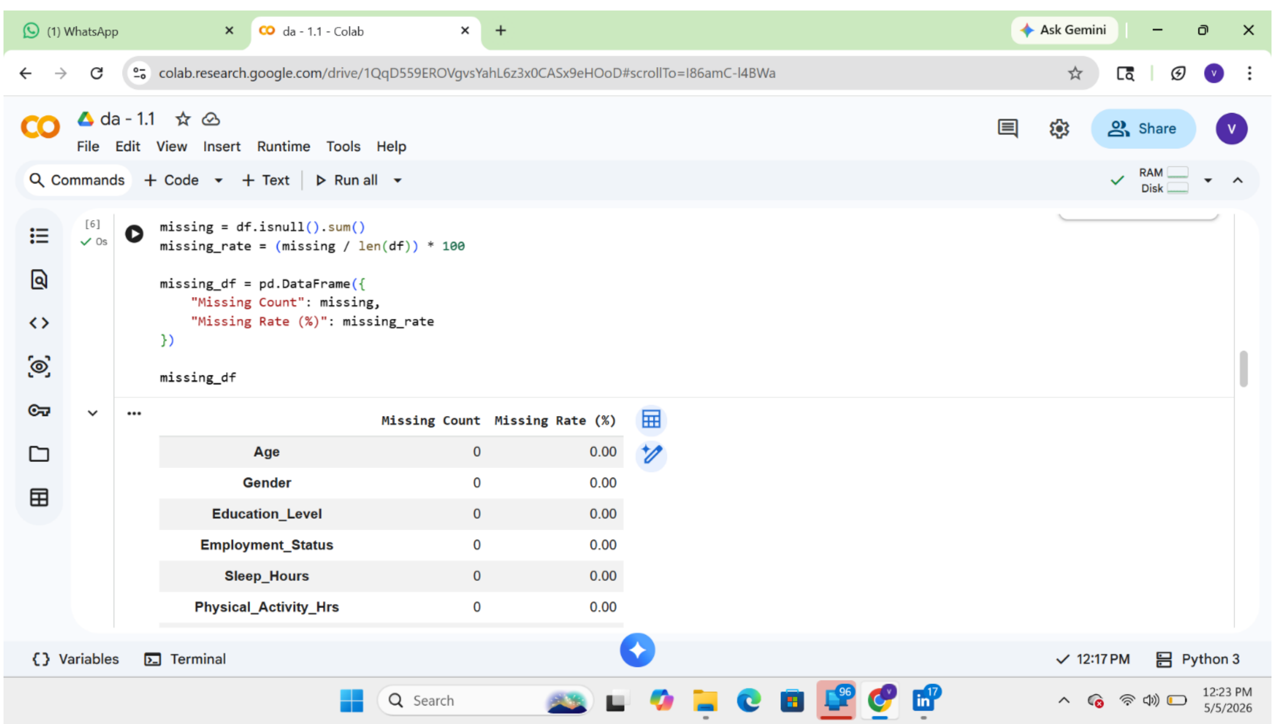


Family_History_Mental_Illness	int64
Chronic_Illnesses	int64
Medication_Use	object
Therapy	int64
Meditation	int64
Substance_Use	object
Financial_Stress	int64
Work_Stress	int64
Self_Esteem_Score	int64
Life_Satisfaction_Score	int64
Loneliness_Score	int64

dtype: object

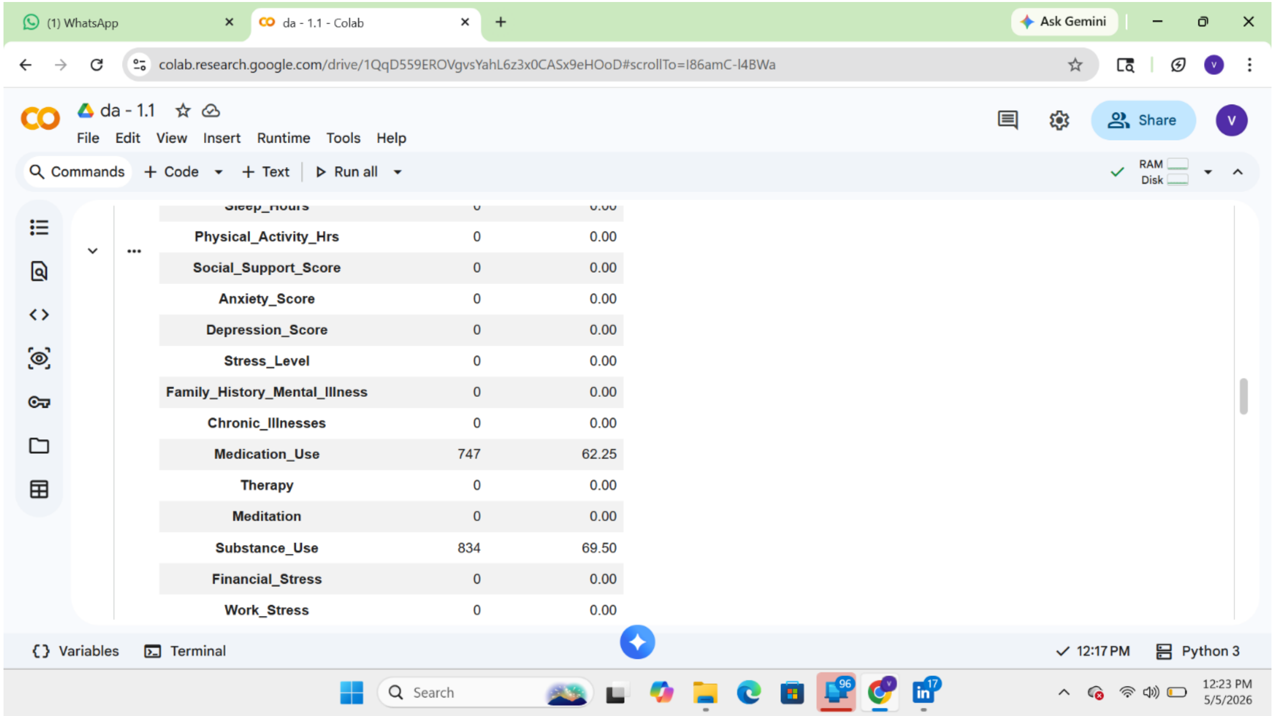
MISSING VALUES:

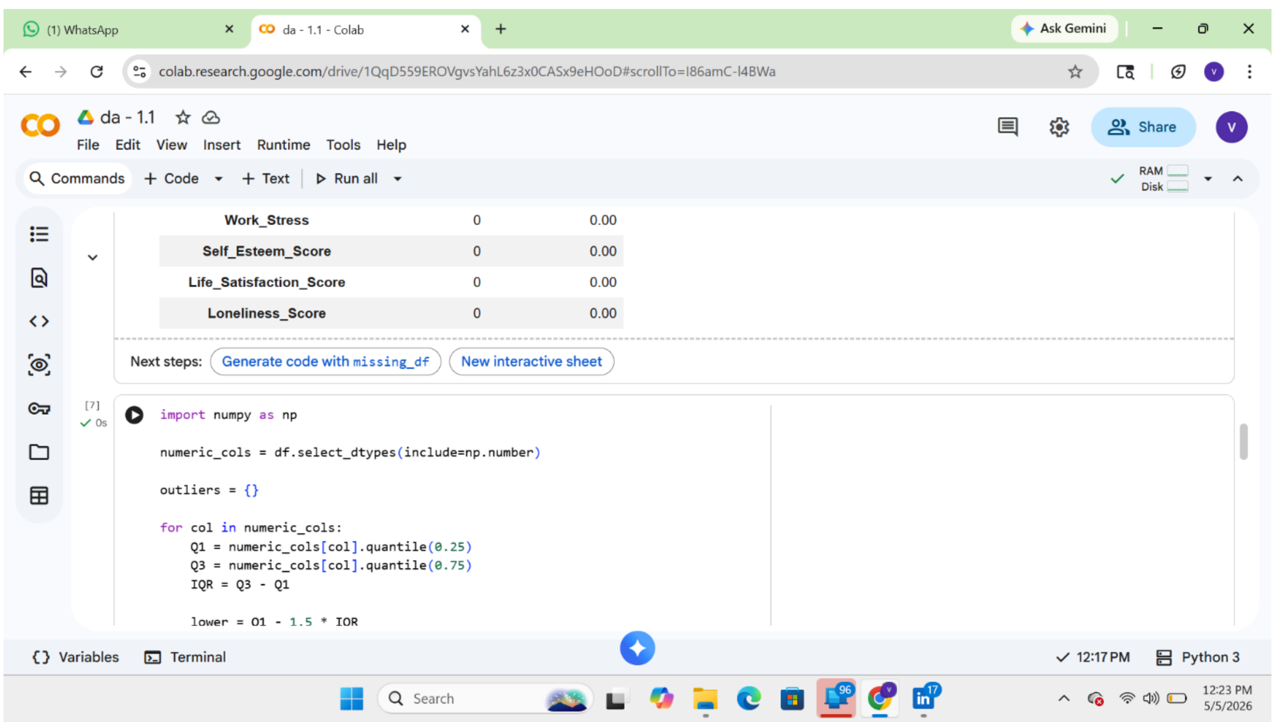
- Detects null or missing values in each column
- Helps in data cleaning decisions (remove/fill values)



```
missing = df.isnull().sum()
missing_rate = (missing / len(df)) * 100

missing_df = pd.DataFrame({
    "Missing Count": missing,
    "Missing Rate (%)": missing_rate
})

missing_df
```

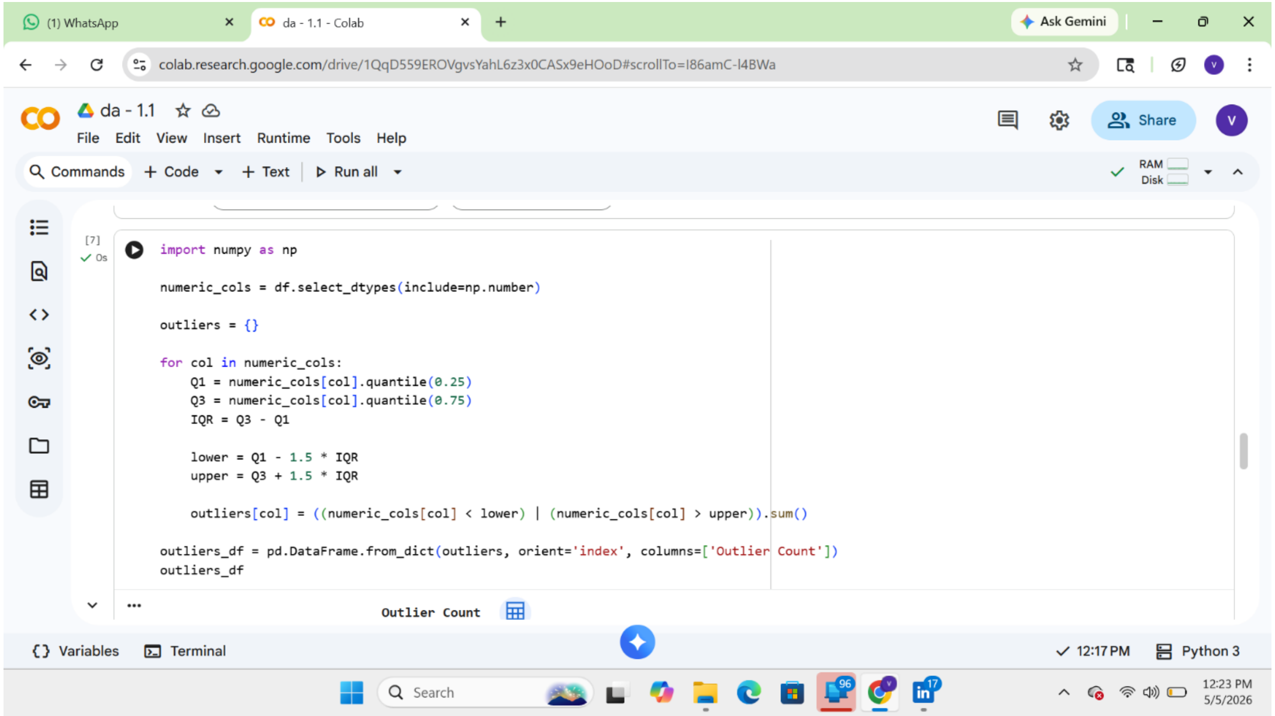
	Missing Count	Missing Rate (%)
Age	0	0.00
Gender	0	0.00
Education_Level	0	0.00
Employment_Status	0	0.00
Sleep_Hours	0	0.00
Physical_Activity_Hrs	0	0.00



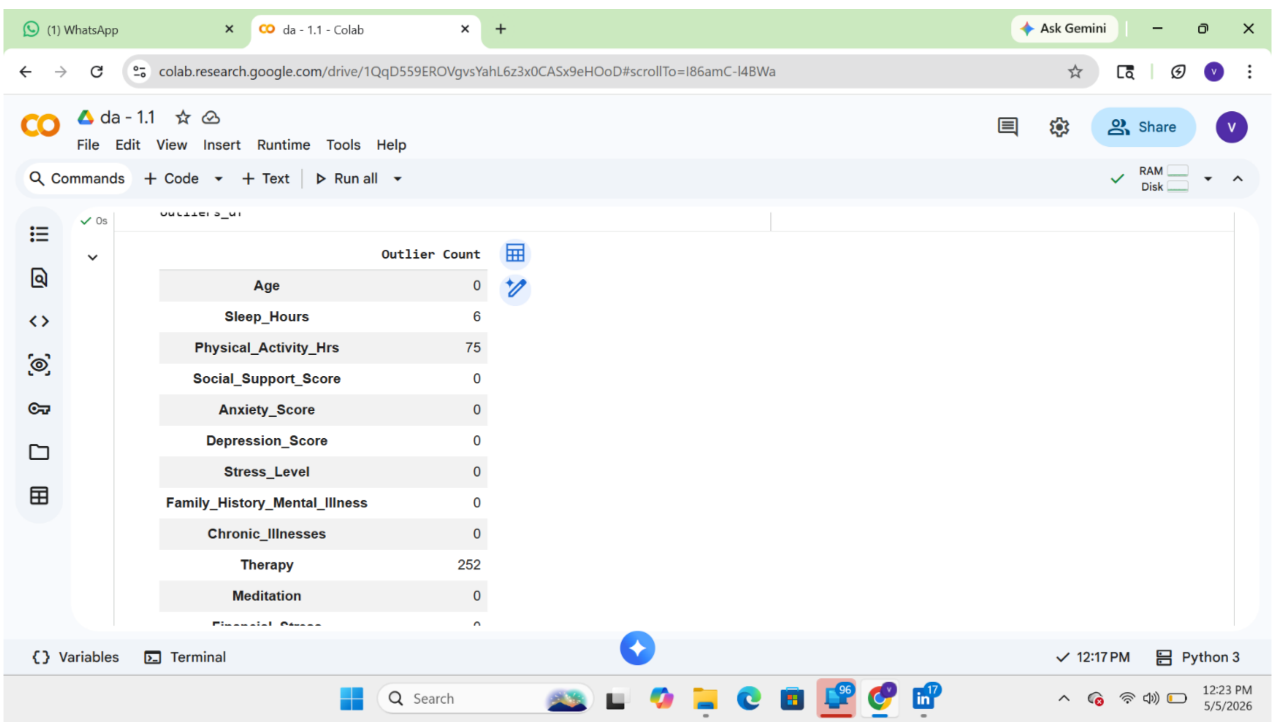


OUTLIER DETECTION:

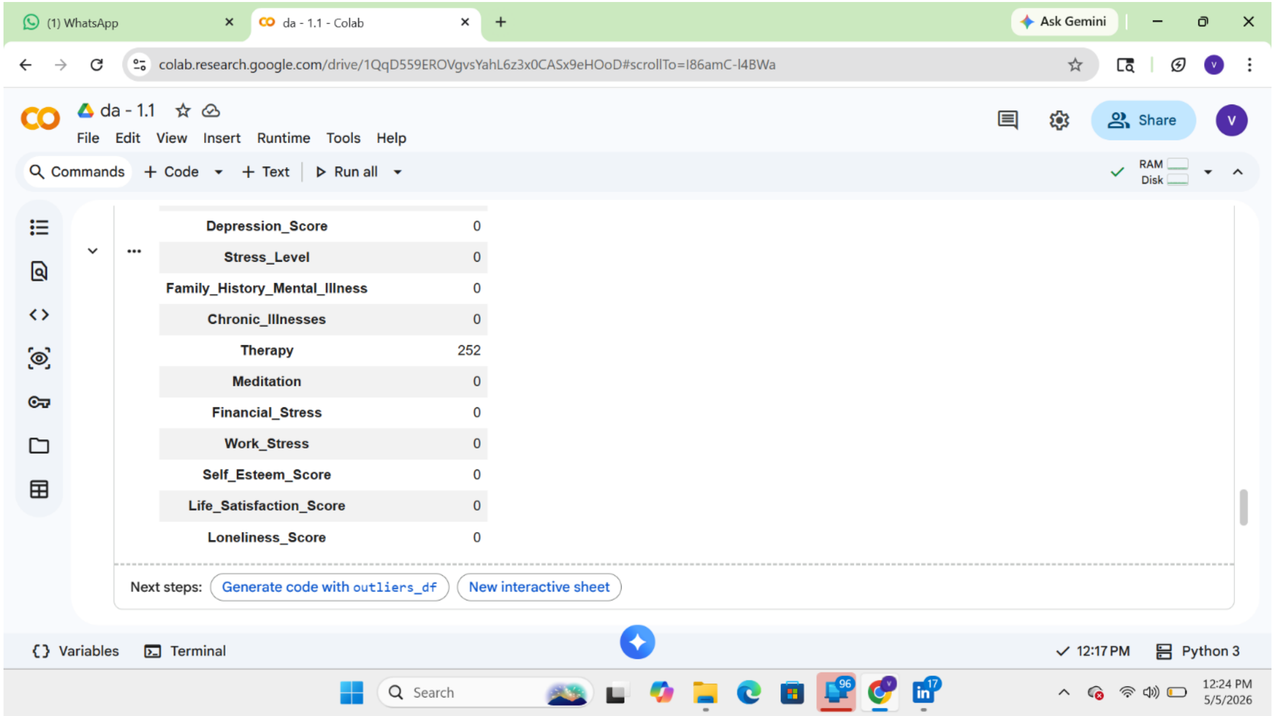
- Identifies extreme values using methods like **IQR (Interquartile Range)**
- Important for improving model accuracy and avoiding bias



```python
import numpy as np

numeric_cols = df.select_dtypes(include=np.number)

outliers = {}

for col in numeric_cols:
    Q1 = numeric_cols[col].quantile(0.25)
    Q3 = numeric_cols[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers[col] = ((numeric_cols[col] < lower) | (numeric_cols[col] > upper)).sum()

outliers_df = pd.DataFrame.from_dict(outliers, orient='index', columns=['Outlier Count'])
outliers_df
```



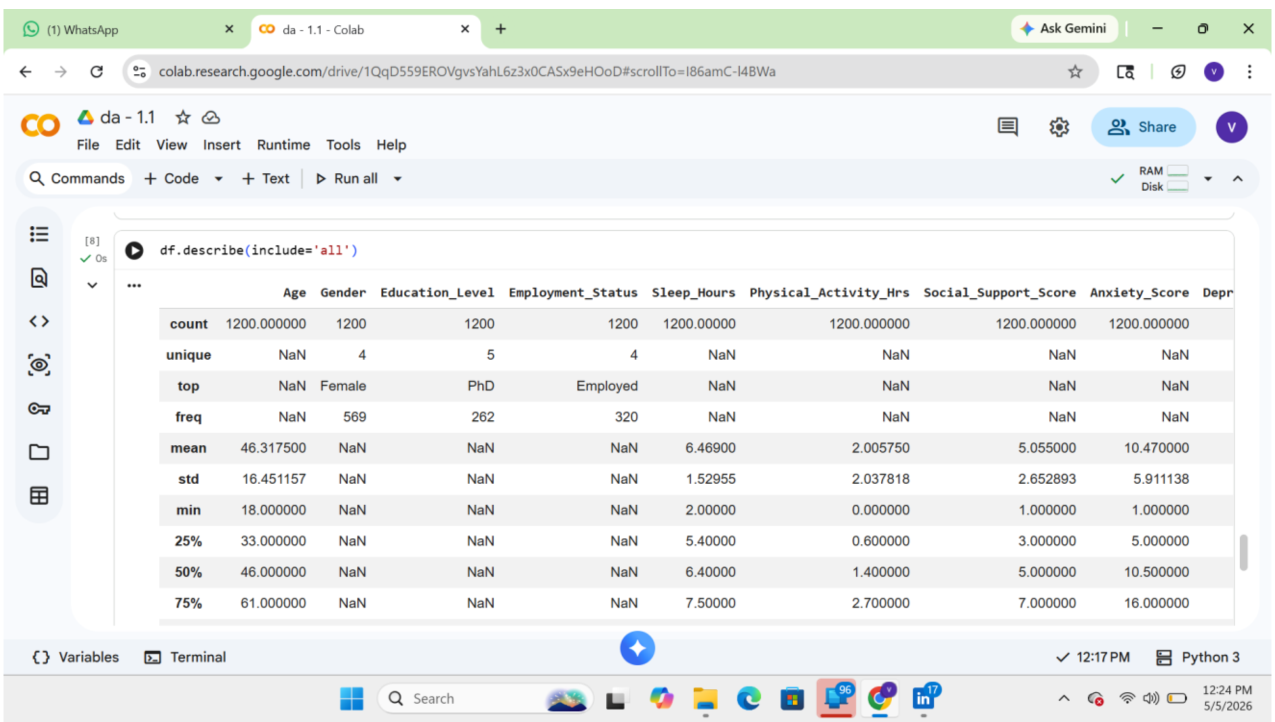
	Outlier Count
Age	0
Sleep_Hours	6
Physical_Activity_Hrs	75
Social_Support_Score	0
Anxiety_Score	0
Depression_Score	0
Stress_Level	0
Family_History_Mental_Illness	0
Chronic_Illnesses	0
Therapy	252
Meditation	0



Depression_Score	0
Stress_Level	0
Family_History_Mental_Illness	0
Chronic_Illnesses	0
Therapy	252
Meditation	0
Financial_Stress	0
Work_Stress	0
Self_Esteem_Score	0
Life_Satisfaction_Score	0
Loneliness_Score	0

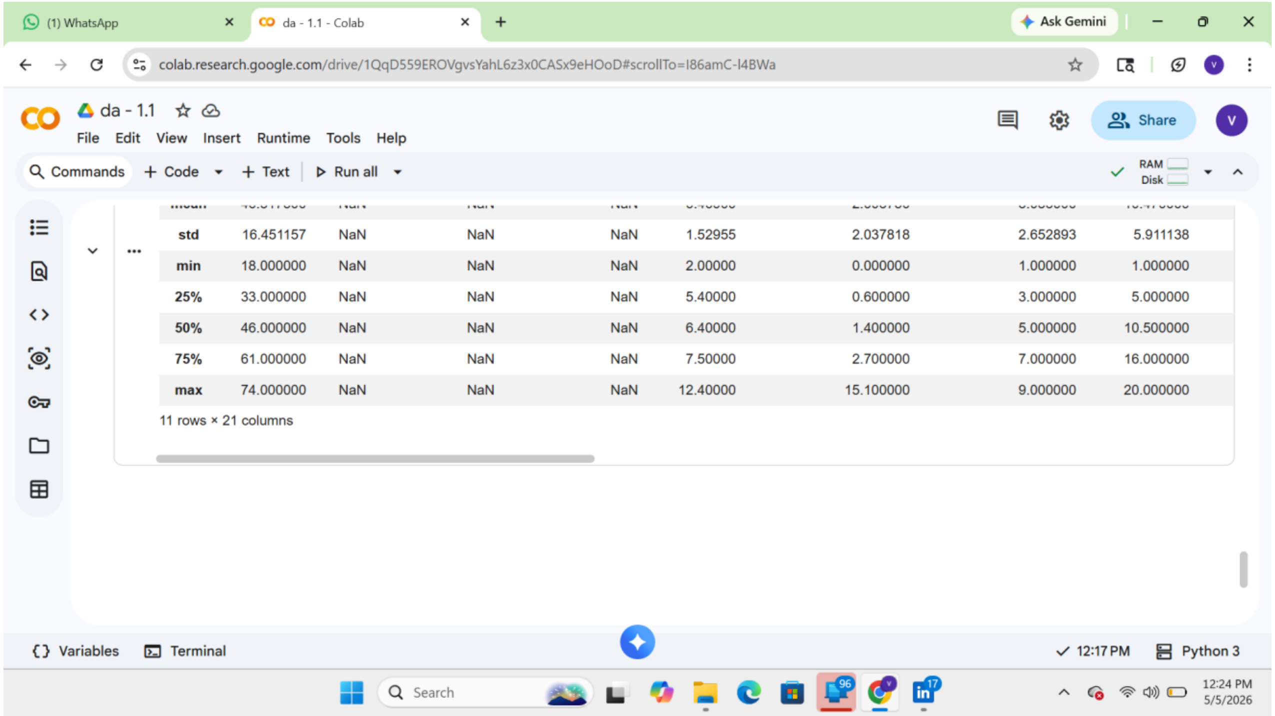
Next steps: Generate code with outliers_df New interactive sheet

Output descriptive stats table:



```
df.describe(include='all')
```

	Age	Gender	Education_Level	Employment_Status	Sleep_Hours	Physical_Activity_Hrs	Social_Support_Score	Anxiety_Score	Depr
count	1200.000000	1200	1200	1200	1200.00000	1200.000000	1200.000000	1200.000000	
unique	NaN	4	5	4	NaN	NaN	NaN	NaN	
top	NaN	Female	PhD	Employed	NaN	NaN	NaN	NaN	
freq	NaN	569	262	320	NaN	NaN	NaN	NaN	
mean	46.317500	NaN	NaN	NaN	6.46900	2.005750	5.055000	10.470000	
std	16.451157	NaN	NaN	NaN	1.52955	2.037818	2.652893	5.911138	
min	18.000000	NaN	NaN	NaN	2.00000	0.000000	1.000000	1.000000	
25%	33.000000	NaN	NaN	NaN	5.40000	0.600000	3.000000	5.000000	
50%	46.000000	NaN	NaN	NaN	6.40000	1.400000	5.000000	10.500000	
75%	61.000000	NaN	NaN	NaN	7.50000	2.700000	7.000000	16.000000	



std	16.451157	NaN	NaN	NaN	1.52955	2.037818	2.652893	5.911138
min	18.000000	NaN	NaN	NaN	2.00000	0.000000	1.000000	1.000000
25%	33.000000	NaN	NaN	NaN	5.40000	0.600000	3.000000	5.000000
50%	46.000000	NaN	NaN	NaN	6.40000	1.400000	5.000000	10.500000
75%	61.000000	NaN	NaN	NaN	7.50000	2.700000	7.000000	16.000000
max	74.000000	NaN	NaN	NaN	12.40000	15.100000	9.000000	20.000000

11 rows × 21 columns

CONCLUSION :

- the dataset contains structured survey data stored in csv format • python with jupyter notebook is used for analysis
- pandas helps in data cleaning and manipulation
- Understanding the dataset clearly
- Detecting data quality issues
- Data issues are identified quickly
- Summary statistics are generated efficiently
- Outliers and missing values are detected accurately